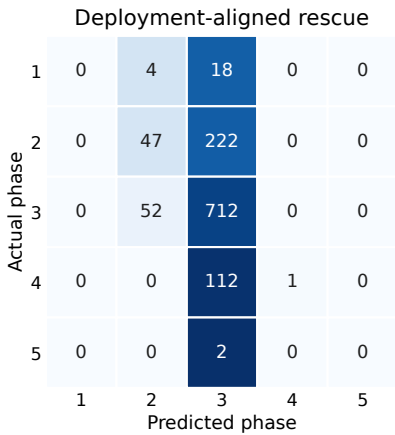
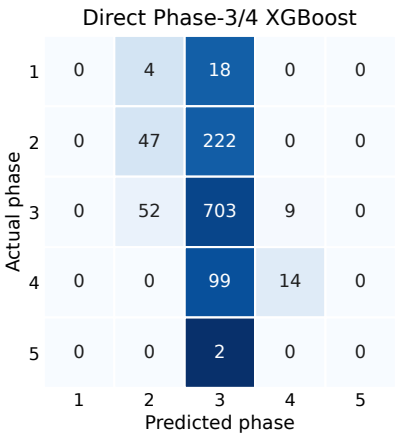
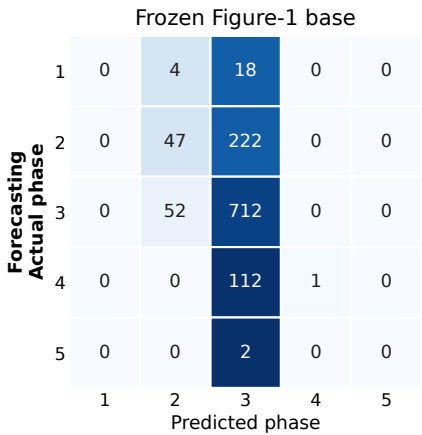


Phase-4 post-classification rescue on the frozen 2022 benchmark

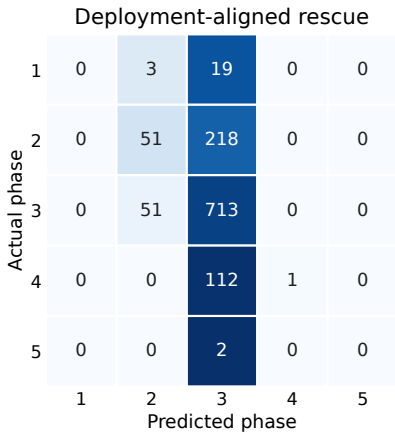
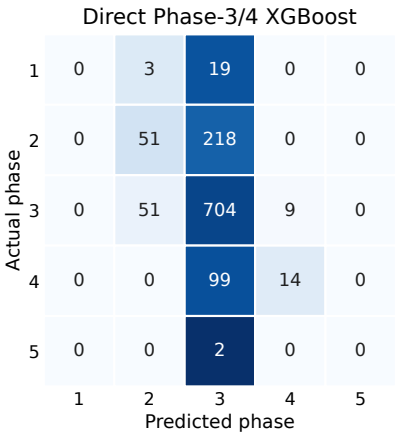
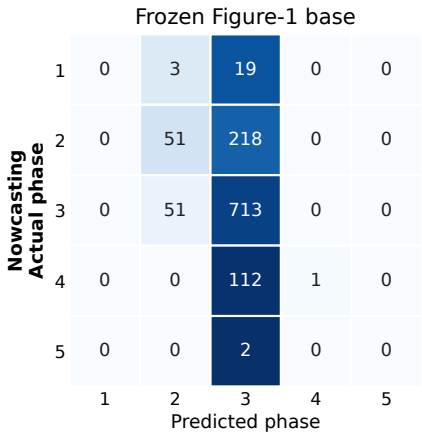
Counts are shown in cells; color is the percentage within each reconstructed actual-phase row (common 0-100% scale).



Forecasting: Main metrics (Δ vs frozen base)

	P3+ precision	P3+ recall	P3+ R ²	5-class accuracy
Frozen base	0.7751 $\Delta +0.0000$	0.9408 $\Delta +0.0000$	0.2490 $\Delta +0.0000$	0.6496 $\Delta +0.0000$
Direct P3/4 XGB	0.7751 $\Delta +0.0000$	0.9408 $\Delta +0.0000$	0.2490 $\Delta +0.0000$	0.6530 $\Delta +0.0034$
Rescue XGB	0.7751 $\Delta +0.0000$	0.9408 $\Delta +0.0000$	0.2490 $\Delta +0.0000$	0.6496 $\Delta +0.0000$

P3+ precision, recall and R² are invariant;
only five-class accuracy may change.



Nowcasting: Main metrics (Δ vs frozen base)

	P3+ precision	P3+ recall	P3+ R ²	5-class accuracy
Frozen base	0.7775 $\Delta +0.0000$	0.9420 $\Delta +0.0000$	0.2528 $\Delta +0.0000$	0.6538 $\Delta +0.0000$
Direct P3/4 XGB	0.7775 $\Delta +0.0000$	0.9420 $\Delta +0.0000$	0.2528 $\Delta +0.0000$	0.6573 $\Delta +0.0034$
Rescue XGB	0.7775 $\Delta +0.0000$	0.9420 $\Delta +0.0000$	0.2528 $\Delta +0.0000$	0.6538 $\Delta +0.0000$

P3+ precision, recall and R² are invariant;
only five-class accuracy may change.

